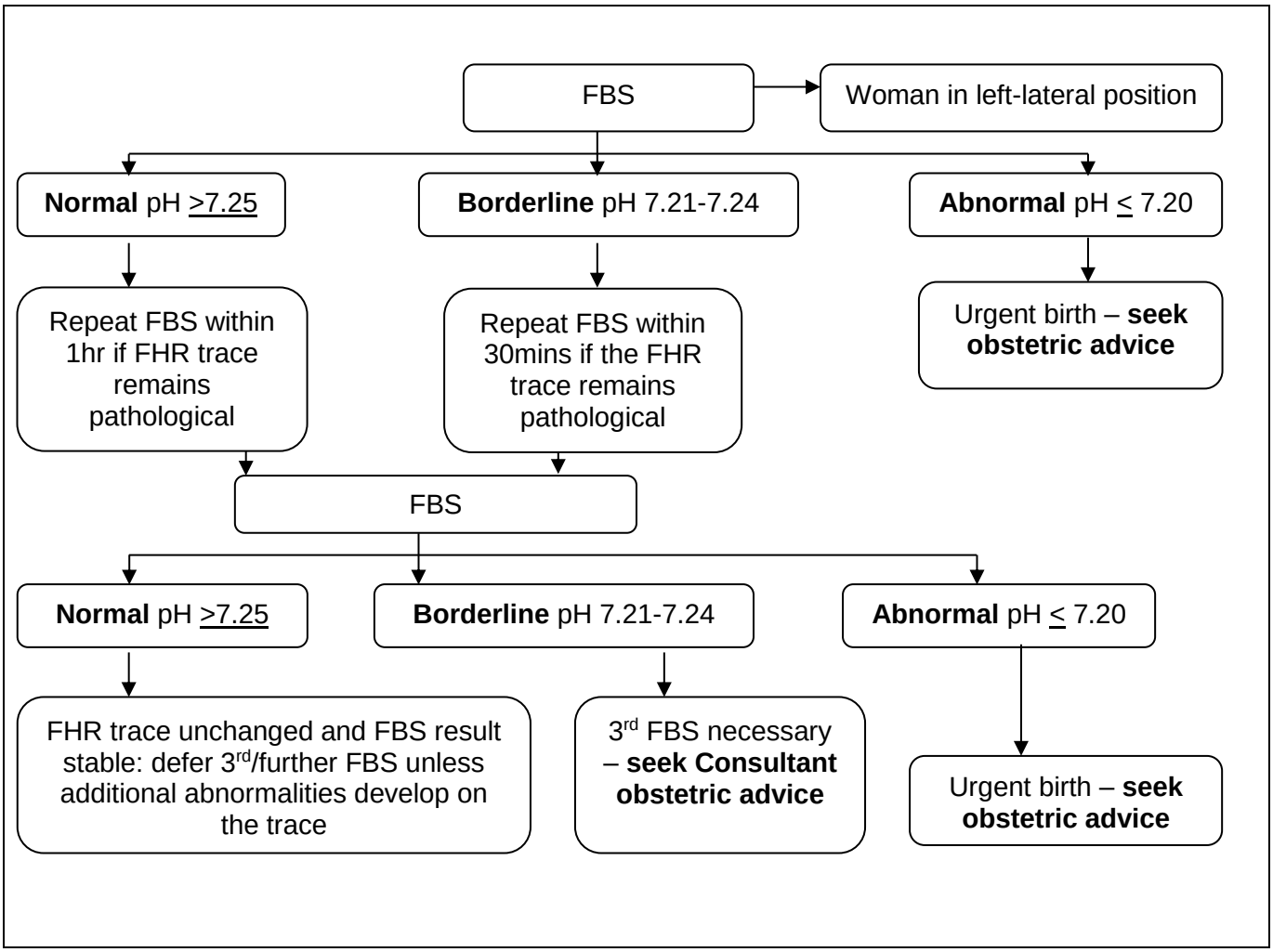
Royal Devon University Healthcare NHS Foundation Trust

Clinic guideline for

Fetal Blood Sampling (FBS) in labour and cord pH

Eastern and Northern Maternity Services

FBS Results Flowchart



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1.0 INTRODUCTION

1.1 Cardiotocography (CTG) is the worldwide method for fetal surveillance during labour. CTG is a screening tool and is used to detect changes in the normal fetal heart rate patterns that may indicate fetal hypoxaemia but it is not able to assess precisely the level of fetal compromise.

1.2 In the event of a pathological CTG, fetal scalp stimulation is used to assess fetal wellbeing. If this leads to an acceleration in the fetal heart rate and a sustained improvement then after a holistic review of the patient, labour can continue.

1.3 There is limited evidence about the role of FBS in intrapartum fetal surveillance. If a clinician feels an FBS will play a role in fetal wellbeing assessment then caution must be taken if suspicious of sepsis or if meconium is present.

1.4 Current national guidance has stated they are unable to make a recommendation about the use of fetal blood sampling due to the lack of evidence.

2.0 PROCEDURE

2.1 FBS is performed by analysis of a capillary blood sample obtained from the fetus.

2.2 It is a sterile procedure and should be carried out with the woman is in the left lateral position (this causes less vena cava compression).

2.3 Ideally, paired samples should be obtained and analysed during any one FBS investigation. The samples should be concordant. Wide discrepancy may indicate sampling or machine error. The lower pH should be used for the purposes of subsequent management.

2.4 The blood gas analysers are kept in the dirty utility room on the Labour ward.

2.5 Do not waste time if the FBS is difficult to obtain or if the CTG indicates immediate delivery. If there is serious concern delivery is preferable.

3.0 INDICATIONS AND CONTRAINDICATIONS FOR FBS

3.1 Indications

Pathological fetal heart trace that hasn't responded to fetal scalp stimulation and fetal pH will assist with ongoing decision making

3.2 Contraindications

- Maternal infection (e.g. HIV, hepatitis viruses, herpes simplex virus)
- Possible or confirmed bleeding disorders (e.g. haemophilia & maternal thrombocytopenia).
- Prematurity (<34 weeks).
- Where there is clear evidence of acute fetal compromise (for example, prolonged deceleration greater than 3 minutes), FBS should not be undertaken and urgent preparations to expedite birth should be made.
- Refusal of maternal consent.
- Non vertex presentation.

- Women in labour with sepsis or suspected sepsis and an abnormal cardiotocograph trace, think about the whole clinical picture and take into account the following before performing any fetal blood sampling and when interpreting the results:
  - The women's preferences
  - Stage and progress of labour
  - Parity
  - Likelihood of chorioaminontis
  - If sepsis continues to be suspected, only repeat fetal blood sampling with caution and in discussion with a consultant obstetrician.

4.0 CLASSIFICATION OF FBS RESULTS AND MANAGEMENT

Fetal blood sample result	**Subsequent action**
(pH)*≥ 7.25	FBS should be repeated within 1hr if the FHR abnormality persists or sooner if CTG deteriorates
7.21-7.24	Repeat FBS within 30 minutes or consider delivery if rapid fall since last sample
≤ 7.20	Delivery indicated
If result unavailable	If CTG concerns remain - deliver baby If scalp stimulation results in acceleration - discuss case with consultant regarding whether to continue or deliver

5.0 SOURCES OF ERROR

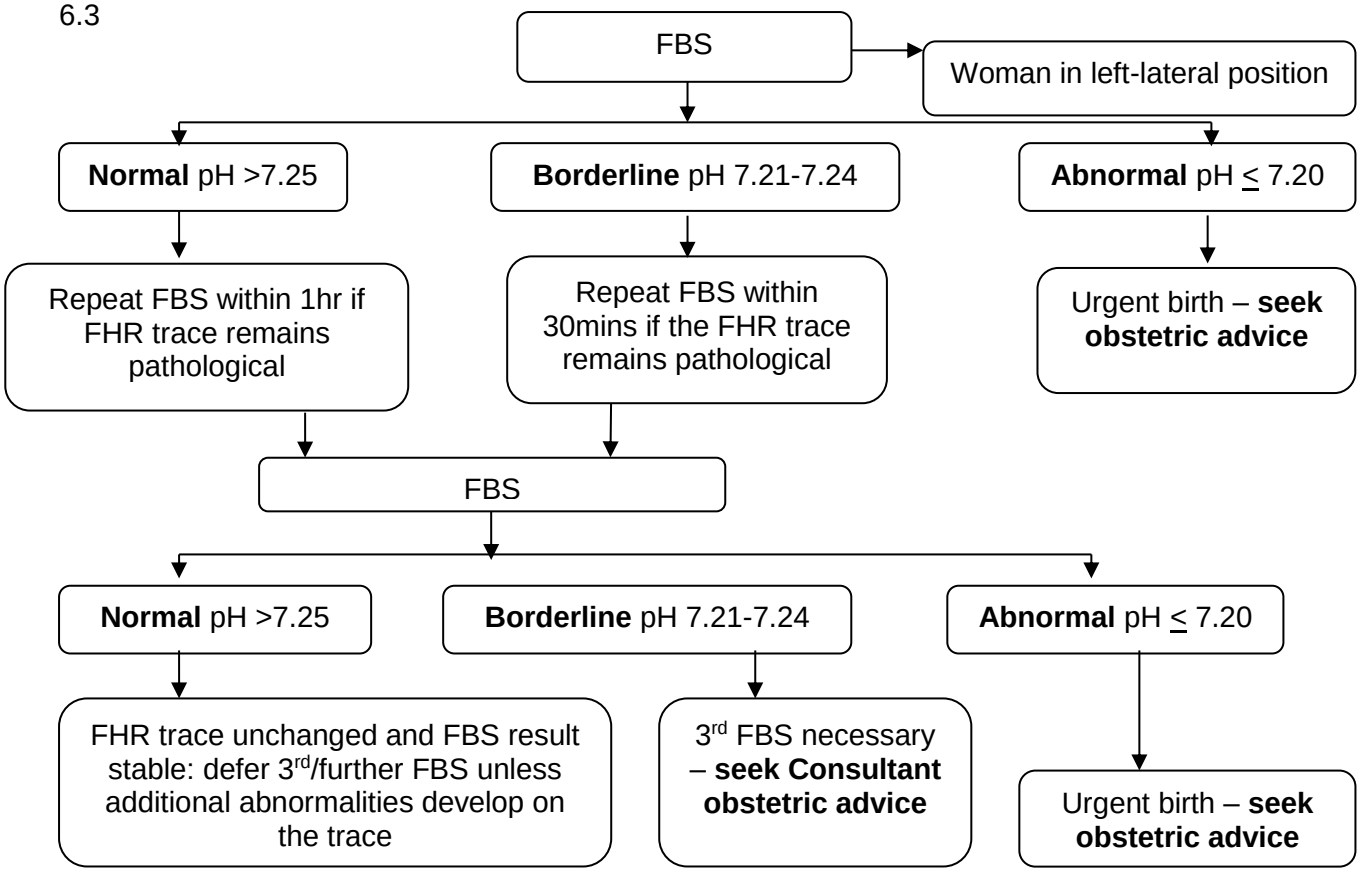
- Contamination with amniotic fluid (↓ Ph value)
- Contamination with meconium (↑ or ↓ Ph value)
- Presence of air bubbles (↑ Ph value)
- Fetal scalp oedema or caput (↓ Ph value)
- Delay in processing (↓ Ph value).

6.0 RESULTS

6.1 Results should be interpreted taking into account the previous pH measurement, the rate of progress in labour and the clinical features.

6.2 If Pathological CTG and result unobtainable, proceed to LSCS / deliver baby. If the CTG improves and reacts to scalp stimulation then discuss case with Consultant and decide whether to continue the labour or expedite the birth in light of the clinical circumstances.

6.3



6.4 FHR traces should be kept for 25 years and archived electronically.

7.0 DOCUMENTATION OF RESULTS

7.1 The FBS results must be documented in the maternal chart via the Labour progress tab and in FBS section. The blood gas analyser will also send a copy of the result to the chart and this can be viewed via the results tab.

8.0 CORD BLOOD ANALYSIS POST DELIVERY

8.1 Paired cord blood samples should be taken **and processed within 30 mins** after the following births:

- If a fetal blood sample has been taken during labour
- If the CTG trace has been Pathological
- Any suspected fetal compromise during the birth
- Shoulder dystocia
- If the baby requires active resuscitation
- Emergency Caesarean Section
- Assisted Vaginal Delivery

8.2 Assessment of the acid-base status of umbilical cord blood at birth provides an objective measure of the fetal response to labour.

8.3 Cord blood gas analysis provides an objective measure of neonatal condition at birth, as opposed to the Apgar score which is subjective, usually assigned retrospectively and is influenced by many factors other than hypoxia during labour.

8.4 The presence of an arterial metabolic acidosis will identify those babies who have experienced significant oxygen deficit prior to delivery, and who therefore may be at risk of neonatal problems such as hypoglycaemia.

8.5 Normal blood gases are essential to defend litigation and acidotic blood gases an essential requirement for a diagnosis of hypoxic ischaemic encephalopathy.

8.6 Without both artery and vein samples you cannot be sure that you have the artery. It is the artery that reflects fetal acid-base balance but is more difficult to get. A single sample cannot be guaranteed to have come from the artery and so is of little value for audit or in a court of law.

8.7 Cord blood gas analysis encourages a physiological-based approach to intrapartum fetal assessment and helps clinicians to retrospectively review and learn from the fetal heart rate trace during labour.

8.8 Paired cord samples provide an objective tool for the audit of intrapartum care and for risk management. The incidence of metabolic acidosis can be used as a quality measure of the standard of care. An arterial cord pH of ≤7.0 should be incident reported on Datix.

8.9 **In the event of a Cord gas result having pH<7.0 or Base Excess ≥16.0, please contact the neonatal team and inform them of the result**

8.10 The paired cord blood results will be uploaded to mum result section of the chart from the gas machine.

8.11 Sampling Technique

- A segment of cord is isolated between two clamps (a 20cm segment is recommended). If this is not done the pH and pCO_2 will change as soon as the baby begins to breathe.
- Heparinised syringes should be used to help prevent clotting.
- The syringe needle should be parallel to the vessel as it is less likely to go through the vessel and into another vessel resulting in a mixed arterio-venous sample.
- **The sample must be analysed within 30 minutes of collection** (RCOG 2007).

9.0 SUPPORTING PARENTS IN CASES OF ACTUAL OR SUSPECTED POOR OUTCOME FOR THE BABY

9.1 The cord pH is useful information to have when counselling parents of babies who encounter problems in the neonatal period.

9.2 An outcome measure is needed that can reliably reflect the condition of the infant at birth and provide an indication of the level of oxygen deprivation ***actually*** sustained during labour. This objective knowledge, can better establish the timing of the cerebral injury (antepartum, intrapartum or postpartum).

9.3 The paediatricians must be informed in cases of actual or suspected poor outcome for the baby.

9.4 The Consultant Paediatrician and Consultant Obstetrician will meet with the parents to discuss the labour, delivery and the possible prognosis for the baby and the details of what was discussed must be documented in the baby's notes or if the meetings have been performed separately then the obstetrician would document in the mother's notes.

10.0 MONITORING COMPLIANCE WITH THIS GUIDELINE

10.1 Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, claims or complaints will be reviewed as per the Trust Policies regarding Incidents, Claims and Complaints, and may result in an audit and/or amendment to the guideline.

10.2 Relevant Policies:

- Incident reporting policy and procedure
- Claims management policy and procedure
- Policy and Procedure for the Management of Complaints, Concerns, Comments and Compliments
- RD&E (2016) Clinical Guideline for abnormal umbilical cord gases – management of well babies

11.0 REFERENCES:

NICE Clinical guideline 190 Intrapartum care: *Care of healthy women and their babies during childbirth.* National Institute for Health and Care Excellence, December 2014.

National Collaborating Centre for Women's and Child Health & their babies during childbirth. *Clinical Guideline 55*. Sept 2007. London.

https://training.k2ms.com

RCOG (2007) Safer Childbirth: Minimum Standards for the Organisation and Delivery of Care in Labour

12.0 PUBLICATION DETAILS

Author of Clinical Guideline	Senior Maternity Matron – Hospital
Directorate/Department responsible for Clinical Guideline	Specialist Services/ CWH/Maternity
Contact details	6682
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Approving body and date approved	Maternity Guideline Group chairs action 12/08/2024
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Expiry date	12/08/2027
Date document becomes live	12/08/2024

APPENDIX 1

Method of obtaining Fetal Blood Sample

- Ensure that the blood gas analyser is in the standby mode.
- Wear universal protective clothing in line with Trust policy.
- Use disposable fetal blood sampling kit.
- Perform a vaginal examination to assess stage of labour and exclude face presentation.
- Position the woman in the left lateral position (may require leg support for the uppermost leg).
- The amnioscope is passed into the posterior vaginal fornix and the obturator is then removed. Use KY jelly for lubrication
- The amnioscope is maneuvered to rest against the fetal head.
- Clean fetal scalp with dental roll, gauze or cotton wool.
- Avoid undue pressure on fetal scalp with amnioscope as this may lead to venous congestion.
- Avoid contamination of sample with Meconium or amniotic fluid as this may lead to a false result.
- Apply a thin film of paraffin wax to fetal scalp.
- A 2mm fetal scalp incision is made with steady pressure of the blade which does not protrude more than 2mm.
- Collect two samples (if possible) of globule of blood in heparinised capillary tubes, allowing the sample to run downhill.
- A 4 cm unbroken column of blood is required. There should be no air bubbles or debris in the sample.
- Seal the capillary tube with the caps provided in the kit before transport to the analyser.
- Transfer the sample without delay to the blood gas analyser and process by a member of staff competent to use the analyser.
- If the blood gas analyser is out of operation and a sample needs to be sent to the laboratory ensure that the technician is awaiting the sample and is aware that an urgent result is required. A porter should be standing by on labour ward to avoid unnecessary delay in transportation.
- Pressure is applied to the incision site with a dry swab until the bleeding stops.
- Reposition the mother once the results are available and inform her of the result and subsequent plan of care.